A TRIALOGUE ACROSS THREE TIME ZONES

Susan Frykberg
School Of Music,
Monash University

Laurie Spiegel
Composer,
unaffiliated

Kenneth Newby
Centre For Culture And Technology,
University Of British Columbia

ABSTRACT

For the past many months three composers, Laurie Spiegel. Kenneth Newby and Susan Frykberg, from New York, Vancouver and Melbourne respectively, have been involved in an internet-based conversation on composition, composing systems, computer-music, spirituality and birds/animals. The conversation was initiated by Melbourne-based composer Susan Frykberg, who wished to discuss a number of issues relating to her PhD at Monash University on spirituality and electronic music. This paper identifies six important themes from the conversations: Reasons for composing; spirituality in music; notation; algorithms; coding and animals. While key themes from our conversation are summarized below, the give and take of the trialogue, which spanned months of an ongoing three-way conversation, is best represented in another form. Therefore a version of the conversation can be seen on Tumblr link. We believe it reveals something of the multi-faceted, evolving and relational nature of the creative musical mind.

1. PARTICIPANTS

Laurie Spiegel is well-known for her use of computers for composing, improvisation and the music software she has written for personal computers. She also does visual art in various media and has published widely on technology in the arts and its socioeconomic effects. She eschews spirituality per se, but is deeply interested in consciousness, cognition and mythology.

Kenneth Newby is a composer-performer, media artist, educator, interaction designer and audio producer. His creative practice explores the use of technology to enable the creation of music, media performances, installations and experiences rich in aural, visual and cultural nuances. His current work is focused on the continued evolution of his computer-assisted composition system, the Flicker Generative Orchestra.

Susan Frykberg is a sound artist and composer who has long incorporated technological, feminist, environmental and spiritual ideas through works for electroacoustics, acoustic instruments, intermedia, chant and improvisation. She has also created music software. She is an active member of several spiritual and religious groups including interfaith organizations and has created liturgies for some of these communities.

2. REASONS FOR COMPOSING

For Laurie, emotion and craft seem to be key. I "craft experiences... They can be experiences of love, wonder, beauty, exhilaration or sadness. The listener is key... Music involves every aspect of being alive, the physical, emotional, intellectual, rational and intuitive parts, and what Susan would call 'spiritual' too. Music by others has helped me many times in my life. I want to do that for others as well as myself. When I find that no music exists that touches, expresses, embodies something intense that I experience inside of myself, I want to create it, for that feeling or imagining not to remain isolated; to express, share, communicate. And I compose out of love of the process, the sense of flow and due to just plain fascination."

For Susan, sound art and composition is a kind of language for commenting on and exploring a wide range of ideas, from women's issues to religion. The language is continually evolving, is sometimes algorithmic and pays particular attention to sonic palettes that have their own kind of 'life.' It incorporates electroacoustic music, soundscape, signal processing, chant and technological innovation, to provide a terrain for exploration and comment on self and other. A specific example is her Audio Birth Project. Fascinated by the experience of giving birth to her son Esha, Susan investigated the birth processes of her sisters and mother. Edited interviews, extrapolated structures, soundscapes, granular synthesis and live instruments were then combined to create four new works.

Kenneth says: "My own practice is grounded in ... research and intuition (finding the sweet spots in that combination).....[and] exploring generative musical poetics. The advantage of working in a generative context is that it enables the composer, through the formulation of a personal theory of music, to effectively explore the infinite design-space of music, to zero in on the richer veins of creative potential. In this sense the

software environment is not based on a concept of artificial intelligence (AI), but rather an amplification of intelligence (IA), that leverages the embodied skills of the composer. The resulting balance of research and intuition produces a novel way of knowing oneself as a composer, performer, and listener."

Crafting listening experiences for others; exploring ideas through developing 'sonic languages'; research, intuition and technological innovation; music for self-knowledge; personal theories of music; sharing and communication of intense inner experience; the amplification of musical intelligence through technology; creating music to help others and generative musical poetics are therefore key concepts. Probably all of them play a part for each of us but with different emphases, depending on the particular compositional project.

3. SPIRITUALITY IN MUSIC

The significance of spirituality is different for each of us, yet we all acknowledge its role in music to a significant degree. Laurie eschews the word for her own practice and prefers to speak of psychology, emotion and myth. Kenneth articulates an evolving form of personal spirituality that integrates a number of concepts from different traditions. Susan is a practicing Catholic, leads Christian meditation and chanting groups and is involved in interfaith dialogue with traditional religions such as Islam and Judaism. Kenneth says "Music as an art of sculpting time for our experience..... [would] seem to be a medium most suited to articulating the spiritual...I've been thinking about the way music allows us to traverse the seemingly infinite space of consciousness. Music is not only resonant with embodied or emotional states, but provokes strong resonance with the flow of consciousness at deeper levels, accessing the subtlety and nuance of the forms of reality we render from the 'blooming, buzzing confusion' of direct experience."

Laurie comments on the distinction between cultures who use music to pursue the ecstatic in spirituality, versus spirituality as quietude, with Sufi music and Gregorian chant as examples respectively. She also notes the role of spirituality in the New York downtown scene in the 70s, saying, "In experimental music here in downtown NY, a lot of the drone and minimalist music is conceptualized as relating to Zen (for example Pauline Oliveros), Tibetan Buddhism (Phil Glass), or the Vedic traditions (LaMonte Young). Music can alter how we experience whatever we're going through, as does meditation and religious belief systems."

Susan is interested in what Jeremy Begbie calls the Great Tradition (Begbi 2007). He argues that music/sound is an analogue for a God-created/infused cosmos, which includes the universe, nature and the human's ethical life. It is also mathematical and geometrical. The Great Tradition begins with Pythagoras and continues through Plato, Augustine and Kepler, via Boethius, who distinguishes between theoretical musical structures describing cosmos, world and seasons, (musica mundana), human nature, (musica humana) and actual music played, (musica instrumentalis).

4. NOTATION

Kenneth comments on how notation tells the performer what to play, but not how to play. He says: "It's good as a mnemonic device - it's a tradition, grid-like, quantized in nature." Kenneth's experience in playing all sorts of notated, un-notated, composed, improvised, and intercultural music has influenced his thinking about algorithmic music: "I've set myself the challenge of creating an orchestra of virtual players and given them the task of expressively interpreting the musical structurings. This has led to the creation of a theory of performance based on a notion of musical information. I includes the metric position of events in a temporal hierarchy, features that can be extracted from a melodic flow, and global states based on higher-level dynamic features such as speed, density, connectedness, loudness, etc. These features are mapped to playing styles that can produce a wide variety of interpretations of a given structure."

Laurie comments on the role of notation in a number of aspects of her musical composition practice. "I do sometimes write musical notations, traditional or graphic, to organize my own thoughts and sounds during the process of composing a work, whether I then fully realize it electronically or give it to other people to play on an instrument. This is notation functioning not as instructions but as an organizational tool for my own musical thinking. Notation is a language in which we can work out structures and go from vague idea to detailed complete thought. I also find pencil and staff paper the freest, most spontaneous way to work with sounds other than live improvisation. It is easier to break out of pattern with a pencil than any computer software I know, and quicker to jot an idea down. It is perhaps the only truly parallelistic, multidimensional human language, accommodating many kinds of structures."

For Susan, the role of notation is constantly evolving and depends on the current project. An electroacoustic piece may require diagrams indicating spectral and temporal material as well as mixing and diffusing instructions. An acoustic piece may use traditional notation and an aleatoric piece may combine text, images, technical information, musical notation and graphics. Two of her recent acoustic works have simply used traditional notation, via the program Sibelius, but most others use a combination of methods. Free-hand drawing of shapes provide structural guidance for early compositional explorations. Rhythmic patterns from Greek and Latin poetic traditions are often used as a base for a variety of accompaniments. Images, sometimes actual art works, are used as guidance for structured improvisation. Fragments of traditional chant, sometimes in its original notation, (neums or pre-neumic notations), are present in some works. Finally, because some of her pieces have a theatrical component, (e.g.

Suffering, premiered at the 2011 ACMC conference in Auckland, New Zealand), use theatrical, technical, textural and lighting instructions as part of the notation.

5. ALGORITHMIC MUSIC

Laurie's works using a variety of her algorithms (Spiegel n.d. A) at Bell Telephone Labs in the 1970s, can be heard on her album The *Expanding Universe* (Spiegel n.d. B). Her program *Music Mouse – An Intelligent Instrument* (Spiegel n.d. C) is well known, having been widely available on several personal computer platforms. She says "I think of algorithmic logic as being like a ladder to let us reach further. Logic lets us take care of decisions we can automate in order to be free to focus on the aspects of music that can't be automated, which we reserve to our intuitive sense of what feels right. Mathematical and logical structures are often beautiful in themselves, but music has to do, at least for me, with the expression and communication, the sharing with others, the embodiment outside of ourselves of both imaginative and emotional subjective experience. We are otherwise alone in our inner experiences, trapped within our minds, experiencing what we do in isolation. Algorithmic logic is an extremely useful tool for that."

Kenneth Newby is a multi-instrumentalist, playing piano, woodwinds, and a variety of gamelan instruments. Yet his interest in algorithmic music makes the computer his primary instrument and his Flicker Generative Orchestra is based on a personal theory of musical structure, its composition and performance. "By a process of introspection—answering the questions around 'how I do it, how I feel it', I seek methods and techniques that can be encoded in a computational form and integrated into an interacting set of procedures for the generation and performance of music. This introspective approach is augmented by a comprehensive study of the literature in theory, cognition, and psychology of music as it is perceived and performed, as well as research into extant approaches to algorithmic composition. I think of this as a grounded personal theory of music. The compositional system is a kind of model of musical mind, or a mirror held up to one's own understanding and abilities of how the form of music works. As Otto Laske (Laske n.d.), one of the pioneers in the field of cognitive musicology characterized it, we engage in a novel form of epistemology or knowing of the musical self when making such models."

While Susan has written programs to compose, her main focus is via sonic categories which somehow have a degree of 'their own life'. Granular synthesis, soundscapes, recorded speech and apps with an AI component, such as SECTOR and Refractions all have, to a greater or lesser degree, specific unique qualities that require detailed attention through repeated listening. They can only be worked with after this process and because of this, the business of composition is probably better thought of as a kind of co–creative endeavour with an existing form of sonic semi-life.

6. ON CODING

For algorithmic composers, coding is an integral part of the practice. Kenneth has described coding as his cognitive yoga – a delightful term that reminds us that the intellect and the spiritual are well-connected. Kenneth says: "Coding is my practice. Can't avoid it! Then again, code can be looked at as a radical development in notation/writing... a notation that has agency, is active... Code that specifies a deterministic musical process, might be read in the same way common music notation is, the reader is able to hear the music represented by the symbols. An interesting difference, of course, is the fact that.. common music [notation] indicates a thing already made, whereas the code specifies its making. Things get really interesting when the code becomes complex enough in its makings that the outcome is not predictable from the processes... and we get emergence—the computational sublime."

Laurie comments that "it's not the code itself that is active. It's a set of instructions functionally comparable to a musical score or a recipe. They all require external agency to produce realized results. In essence, an algorithm is a general statement describing a process and producing results. It is a shorthand for many specifics, an automation of process. We can concisely specify many details by generative algorithms, and to the extent that we can figure out our own musical decision-making processes as composers, we can automate what we do ourselves in our minds almost by musical reflex...We are free to feel more, go deeper, concentrate more intently on what is not automatic or a reflex in our minds, and to see the whole instead of being bogging down in detail. Often though, we find something unexpected that is wonderful... It's a form of introspection to try to capture in logic what might make good music...and to generate what sounds right to our ears. It's an exciting learning experience as well as a way to increase our musical capabilities."

6. ON ANIMALS

Each of us have a particular affinity to animals – Laurie to 'underdog species' especially pigeons, (Feisst n.d.) as well as a lifelong succession of pets; Kenneth to his 'wild companions' and Susan to the large number of pets she takes care of in the greater Melbourne area in her role as professional house-sitter.

Laurie says: "Animals live much more entirely in the moment than we do. And they don't let language and all the abstractions which complicates our awareness, interfere with their being in the moment here and now. That's not to say that they don't have memories, fears, wishes, the awareness of futures possible and past. But they are much more in the moment I think, and they can concentrate on what or who is right in front of them more purely and intensely than us distracted beings...So we might think of animals as natural meditators."

Kenneth speaks particularly fondly about wild animals – frogs, ravens, hummingbirds, a snake, and adds: "when we attune ourselves to that 'in the moment-

ness' with them we're gifted with some of their way of being/becoming in the world. It always feels like a great honour to have the trust of a free (non-captive) animal. In these interactions we get the opportunity to share the experience of being more than the sum of the bodily functions encapsulated by our skin, as well as grasping at different ways of being in the world. I try to know myself better through my interactions with them."

Susan also comments on the spiritual relationships between herself and the animals she takes care of as a professional house-sitter. They provide her with companionship and an 'in the momentness' close to the quality of contemplation. Many of the animals are regulars and there is a sense, it seems from both sides, of spending time again with 'old friends'. For new animals, there is always a time of adjustment: "We take a while to co-habit on a psychic level."

7. IN SUMMARY

This paper briefly identifies six themes from our three-way conversation: reasons for composition, spirituality and music, notation, algorithms, coding and animals. Key themes from our conversation are summarized here, but more of the give and take of the ongoing trialogue the can be seen on the Tumblr link. We believe that the conversational form is of particular interest to composers because it indicates the relationality and multi-faceted, evolving and relational qualities of the creative, contemporary musical mind.

8. REFERENCES

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